

April 30, 2026

Dear *Lifecycle Journal* editors,

Thank you for organizing evaluations of our research plan submission, “Evaluating the Claim that Preregistration and Registered Reports Restrict Exploratory Research.” We received assessments on our proposal from three evaluation services: MetaROR, PaperWizard, and ReviewerZero. We respond to each evaluation below, in that order. The comments provided were thoughtful, helpful, and have greatly improved the research plan.

We have enclosed a clean PDF as well as a Word doc with all changes tracked.

As a preface to our comments, we want to highlight the context of this research project. We submitted this research plan as part of *Lifecycle Journal*’s pilot program that is focused on assessing the usability and feasibility of the service. This program has strict deadlines. The research plan had to be submitted by Dec 1, 2025, with a 60-day evaluation window, and the final outcomes report must be submitted by Sep 1, 2026. This is a very tight timeline, and so we proposed a small project that would be achievable within those constraints. The tight timeline was exacerbated by an extended evaluation process—instead of 60 days from submission, we had all evaluations in hand approximately 135 days after submission.

We offer this explanation up front as a partial rationale for why we declined to take many reviewer suggestions related to expanding the scope of the project. We simply do not have time or capacity to do so. Although this is never a sufficient scientific explanation, we maintain that our planned project, following the revisions based on the helpful comments, will nevertheless provide informative data, even if on a narrow question. Moreover, the dataset of articles we are assembling will have high reuse potential for metascientists, and some of the suggested expansions could certainly be pursued by us or others down the line.

Finally, following submission of this revised research plan we will formally register the project and begin, and thus the submitted version is the final version of the research plan. Accordingly, whereas we are happy to receive additional comments on the revised plan, we will unfortunately not be able to take any of them into consideration as part of the preregistered plan (we could, of course, incorporate some of them as unregistered aspects).

Thank you again for the time and input.

Cordially,

Moin Syed, Caroline Armstrong, Emily Chan, and Abby Person

MetaROR

Editorial Assessment

I much appreciated the detailed explanations for the decisions made in the design, which made it much easier to arrive at an informed assessment. That level of detail also seems to have helped the reviewers provide the elaborate feedback they did. Both reviewers agree that the manuscript takes up an important question, but both also think the current proposal does not yet support the claims it wants to make. In particular, the reviewers think that the study can speak to how exploratory analyses are reported or labeled in published papers, but not to whether preregistration and Registered Reports actually reduce exploratory research itself, or diminish creativity, novelty, and serendipity. They also raise related methodological concerns: the definition of exploratory research is still too loose for reliable coding, the coding plan and statistical analyses are not yet sufficiently specified, and some design choices (coding at the article rather than study level, ignoring differences in preregistration quality, and not accounting for journal-specific reporting requirements) make the eventual interpretation of the results difficult. The reviewers are largely aligned on these central points. However, reviewer 1 puts more weight on the broader framing of the debate and on the possibility that the study is really picking up journal policy rather than research practice, while Reviewer 2 focuses more on the need for a tighter protocol and a clearer analysis plan. The manuscript is well written, but some of the language in the abstract and Methods section is imprecise, and that lack of precision contributes to the larger conceptual problem.

Author response: Thank you, we are glad to hear you appreciated the detail. We are grateful for the reviewers' detailed and helpful feedback, and provide full responses below following the reviewer comments.

Recommendations from the editor

The main revision needed is a clearer match between the paper's framing and what the proposed design can actually show. At the moment, the study seems able to examine the explicit reporting of exploratory analyses but not the actual frequency or value of exploratory research. The abstract and introduction should reflect that. Additionally, it should not be implied that the study will test whether preregistration restricts creativity, discovery, or serendipity unless those outcomes are going to be measured directly.

Author response: We have now clarified the framing and the scope of the project throughout the text, making it clear that we are analyzing reports of exploratory research vs. the actual practice, per se. We have also added the following to the paper, explaining why we are taking this approach and how there is not really any good alternative:

“A major challenge to drawing any conclusions about the frequency of exploratory research between registered and non-registered articles is that the ability to do so is entirely dependent on researchers' reporting. Given the widespread prevalence of HARKing, in which exploratory findings are reframed and presented as confirmatory, it is simply not possible to derive an accurate estimate of how often exploratory research is occurring when there are no reporting restrictions in place (Wagenmakers et al., 2018). Thus, when making such comparisons, examining the reporting practices in articles is the best available indicator. Although reports are an imperfect proxy for actual practice, they remain highly relevant to the debate about the status

of exploration in registered articles, and specifically address the claims that registration would restrict exploratory research. That is, if registration does in fact restrict exploration, there should be low levels of exploration reported in research articles.”

The manuscript would benefit from an explicit coding protocol that spells out how raters will identify exploratory analyses and what counts as sufficient evidence that an analysis was unplanned. It is difficult to assess whether this important assessment is valid based on the text in the manuscript alone. Both reviewers already indicated some concerns about this validity. Moreover, the coding plan seems to focus mainly on the Introduction, Results/Data Analysis, and Discussion, but in psychology papers, analytic decisions are often not presented in a separate data analysis section but folded into the Methods section. As a practical adjustment, I think it would make sense to also code those parts of the Methods section that describe the analysis plan, even when they are not explicitly labeled as such.

Author response: The full coding protocol is now included in the OSF page. And yes, we agree the descriptions of the analytic plan sometimes appear in the Method section, and so we will review those sections as well.

I would also encourage the authors to reconsider the decision to code at the article level rather than the study level, especially in multi-study papers that may mix preregistered and non-preregistered components. That choice currently weakens the inference the paper wants to make. As reviewer #2 suggests, it may even be informative to do a within-paper analysis, comparing preregistered and non-preregistered studies that way.

Author response: Indeed, this decision evoked strong reactions from the reviewers. We continue to believe that our rationale for coding at the article level is sound, but it is obvious that others do not see it that way. Accordingly, we have revised our inclusion criteria to only consider articles where all studies are preregistered. We will then conduct exploratory within-article examinations of the articles that consist of registered and non-registered studies. Our preliminary article screening has indicated that mixed Registered Reports constitute only a small number of articles, so there are unlikely enough to conduct any hypothesis tests. Excluding them reduces heterogeneity in the focal set, while the exploratory analysis could provide some interesting insights.

A final issue, not mentioned by the reviewers, is the sampling frame. Because the study starts from journals known to publish Registered Reports, the final sample may end up drawing disproportionately from journals that are already more committed to transparency reforms. That does not make the design inappropriate, but it does limit how broadly the findings can be interpreted. One possible way to reduce this problem would be to relax the requirement that companion articles come from the same journal as the Registered Report, although I recognize that doing so also has its downsides.

Author response: We agree that this is a potential limitation. Selecting journals/articles and matches is difficult, and always involves some trade-offs. On balance, we believe it is critical to select matches from the same journals and timeframes, given the variations in journal policies (as

identified by the reviewers). As a way to check the implications on our decision, we have added a small sample of matched traditional articles from journals that do not publish Registered Reports. Doing so will provide some indication of the representativeness of the journals selected.

Review #1

Major comments

I am not fully convinced by the manuscript's framing of its contribution to the debate over whether preregistration hinders exploratory research. The authors position the study as informing this broader discussion, but the proposed design seems able to support conclusions only about reporting practices but not research practice itself. This is important because critiques of preregistration typically concern not just whether exploratory work is conducted (or reported), but whether preregistration constrains creativity and novelty and ultimately hinders the progress of science. Assessing disclosure about exploratory research without also examining these related dimensions therefore seems incomplete.

Author response: As noted in our response to the Editor, we are now clearer about the scope and about why we focus on reporting practices. We are now clearer in the paper that we are narrowly focused on reports of exploratory research.

By the way, the reverse inference is also problematic: even if non-preregistered articles disclose exploratory analyses, this does not mean that the remaining analyses are genuinely confirmatory. This issue is, at present, not addressed in the manuscript.

Author response: Yes, we agree, and made no claims about prevalence of confirmatory analyses. This continues to be the case in the revision.

More broadly, the proposed design appears to conflate research practice with reporting standards. Registered Reports may show a higher frequency of exploratory analyses simply because journals often require such analyses to be reported in a dedicated section, whereas comparable guidance may not exist for regular articles. For the focal sample and the most common journals, it would therefore be important to discuss and, if possible, analyze the relevant reporting guidelines, while taking into account that these policies may have changed over time. For example, the submission guidelines for Psychological Science state for Registered Reports: "It is reasonable that authors may wish to include additional analyses that were not included in the registered submission. For instance, a new analytic approach might become available between IPA and Stage 2 review, or a particularly interesting and unexpected finding may emerge. Such analyses are admissible but must be clearly justified in the text, appropriately caveated, and reported in a separate section of the Results titled 'Exploratory analyses'." As far as I can tell, this recommendation does not appear in the same way for regular manuscripts. This raises the question whether the study measures researcher behavior (bottom-up research practice), or simply adherence to journal policy (top-down reporting requirements)? The authors should account for

differences across journals and reporting guidelines (also in their statistical model), as any differences across article types will be difficult to interpret otherwise.

Author response: Thank you for raising this important point. We agree about the expectation of reporting for Registered Reports, which is one reason we are including preregistered articles. It is, unfortunately, not feasible to examine how journal policies have changed over time and to map that to specific articles, as journal policies are updated often and rarely archived. As a way to partially address this concern, we will code for current journal policies, as now described in the manuscript, and conduct exploratory analyses on their potential relation. Moreover, in response to the comments from PaperWizard, all analyses will be clustered by journal, so at least we can account for journal-level variation.

For the same reason, the manuscript would be more convincing if it also studied the hypothesized consequences of discouraging exploratory research. The Introduction notes the criticism that preregistration may hinder creativity and reduce novelty, and these are precisely the kinds of outcomes assessed by Soderberg et al. (2021) and O'Mahony. The study would therefore be stronger, and better aligned with the concern raised in the Introduction, if it assessed not only whether exploratory analyses are disclosed, but also how novel, creative, rigorous, and transparent the resulting research is.

Author response: We agree that doing this would be informative. As noted at the outset of this letter, this current project is small scale, focused, and on a very tight external deadline. We believe that generating data related to the narrow claim about registration is still useful. Moreover, there are not good measures available to assess creativity and novelty in these kinds of articles. Soderberg et al. relied on single-item assessments with no descriptive text, and O'Mahony did not assess creativity and novelty. Thus, we will indicate this as a limitation and a useful direction for follow-up work.

Overall, the paper currently reads as a replication of one of O'Mahony's hypotheses, with the extension to include regular preregistrations. This should be stated more clearly in the Abstract and Introduction. Although such a replication may well be worthwhile, the present proposal would, in my view, require substantial methodological improvement to justify publication. For instance, O'Mahony distinguished exploratory analyses reported in the paper in relation to the studies actually conducted, whereas the justification offered here for coding at the paper level is not yet convincing. In addition, O'Mahony coded several dimensions that seem at least as important as reporting standards alone, including methodological rigour, novelty, creativity, and transparency. At present, the manuscript appears to treat the presence of reported exploratory analyses as a proxy for these broader constructs without measuring them directly, even though they could in principle be incorporated into the coding scheme. If, instead, the authors are primarily interested in reporting practices across publication formats rather than the substantive quality of the research, then it may be worth broadening the coding scheme to include other reporting-related features as well, such as clarity of hypothesis specification, transparency regarding data, materials, and code, etc. In the latter case, this would require a reframing of the Introduction to reflect this shift in focus.

Author response: We agree that these are all good ideas, but seek to keep the scope limited to the narrow question of interest. We have now more clearly indicated that the study is a replication and extension of O'Mahony. Contrary to the reviewer's claims, O'Mahony did not assess creativity and novelty.

Finally, I do not find the rationale for disregarding preregistration quality convincing. As the authors note, preregistrations can vary substantially in quality, and this variation may plausibly moderate how rigorously exploratory analyses are reported. Treating all preregistrations as equivalent therefore seems difficult to justify, particularly in light of existing evidence that preregistrations differ considerably from one another.

Author response: We agree that examining the contents of the preregistrations would be informative, but we have decided against doing so for two primary reasons. First, it is simply not feasible to do so within the constraints of the project. Second, it is not entirely clear how to think about registration quality in relation to our research questions. As this reviewer and others have noted, we are examining reporting in the articles, and that is useful regardless of registration quality. Moreover, undisclosed deviations from registrations do not really represent exploratory research in its intended sense, but rather are much more likely to represent explorations that were reframed as confirmatory. Although this is of course a type of exploration, it represents something qualitatively different from direct reports of exploratory work.

Minor comments:

It is currently somewhat unclear how the dependent variable, that is the presence of exploratory research, will be operationalized. O'Mahony used a proportion score based on the exploratory analyses documented in each paper and the proportion of analyses where this distinction was clearly made. My understanding of the present proposal is that each article will simply be coded for whether exploratory research is reported, which would make the outcome variable binary. The authors should clarify whether this interpretation is correct.

Author response: We have clarified the coding process in the manuscript and in the appended protocol. It is true that O'Mahony used a proportion score, but the analysis reported just before that used a binary presence rating such as the one we had proposed. As our focus is only on exploratory research, we only include the presence ratings.

As written, the claim that “the function that preregistration is meant to serve is not always clear or consistent” seems too broad if it is mainly based on a preprint by Lakens (2019). Much of the literature describes the function of preregistration quite consistently as distinguishing confirmatory from exploratory research. For example, Nosek et al. (2018) state in the abstract that preregistration is “an effective solution” because it “distinguishes analyses and outcomes that result from predictions from those that result from postdictions.” Similarly, Munafò et al. (2017) write that preregistration “makes clear the distinction between data-independent confirmatory research ... and data-contingent exploratory research.” Parsons et al. (2022) likewise define preregistration as a practice that “aims to clearly distinguish confirmatory from exploratory research.” The authors

may therefore wish either to qualify this claim and/or clearly attribute it to the single source (i.e., Lakens, 2019) who claims that there is disagreement about the function of preregistration.

Author response: We agree, as noted in the paper, that the exploratory/confirmatory distinction is a primary rationale for preregistration. But it is well-documented that it is not the only rationale. This is well covered in Lakens (2019) as well as Syed (2024), the latter of which briefly summarizes these different rationales:

“One of the challenges of understanding preregistration—and the criticisms of it—are that there are different rationales for why researchers should do it. These include clearly distinguishing between what decisions were made prior to seeing the data (“confirmatory” analyses) from what decisions were made after seeing the data (“exploratory” analyses), and preventing the latter as being framed as the former in a research report ; reducing the prevalence of undisclosed data-dependent decision-making ; evaluating the severity of a test ; and serving as formal documentation of the study design and analysis plan.”

Additionally, we do not see why a single source, which we do not rely on anyway, is a problem, nor is relying on a preprint (as it happens, this was an error in our Zotero library; the paper is indeed published, and this has been updated). Accordingly, we have not made any changes in response to this comment.

The authors may wish to discuss additional relevant literature alongside the thesis by O’Mahony. Although Soderberg et al. (2021; <https://doi.org/10.1038/s41562-021-01142-4>) did not examine exploratory work directly, they did evaluate the novelty, creativity, and quality of Registered Reports relative to standard publishing models, and these findings seem directly relevant here. Wagenmakers et al. (2018; <https://doi.org/10.1177/1745691618771357>) may also warrant discussion, as it provides a useful historical overview of the relationship between creativity and verification.

Author response: Thank you, we have added reference to these articles.

The OSF materials appear to be incomplete and do not currently include the analysis script.

Author response: Thank you, the scripts are now available as indicated in the manuscript.

The abstract contains an incomplete sentence (“The purpose of the present study is to find out.”) and should be revised to state the study aim clearly.

Author response: We have revised the sentence for clarity.

When preregistration is first mentioned in the Introduction, the authors may wish to note that it was proposed as early as 2012 and cite Wagenmakers et al. 2012; <https://doi.org/10.1177/1745691612463078>). Similarly, when introducing Registered

Reports, Chambers (2013; <https://doi.org/10.1016/j.cortex.2012.12.016>) would be a more appropriate reference than Besançon et al. (2021).

Author response: Both of those articles were already cited very early on in the Introduction. We realized that the citation placement of Besançon et al. (2021) was a bit misleading, so have made edits to clarify.

Review #2

It was my pleasure to read and review this registered report.

After reading the report, I unfortunately come to the recommendation not to publish this work (in its current state). The criterion I use is the one that is described in the paper: “he Stage 1 manuscript and review process, addressing the question: will this study produce worthwhile knowledge, regardless of the results?” In my opinion the study, as currently planned and described, will not produce worthwhile knowledge. I provide my reasoning below under ‘major comments’.

However, I hope the authors are willing to adapt their proposed work, as I believe it may produce some worthwhile knowledge when (strongly) adapted.

I also provide some minor comments.

Author response: We appreciate the helpful feedback!

Major comments

1. Claim and framing. Your abstract and your paper state that your research is about the frequency and value of exploratory research:

(a) Most importantly, your research is NOT about the frequency of exploratory research. It currently is about the the reader being able to identify whether research is exploratory or not. This is something very different. Research that is not preregistered does not identify or label research as exploratory or confirmatory, as this distinction is not relevant or cannot be made – I would say almost all of this research is exploratory as ideas and hypotheses are either not formulated before the research or adapted during the research. My idea is supported by the results of the dissertation of O’Mahony, which did or could not identify exploratory research in non-preregistered research.

Hence, I think the aim needs to be reframed into examining the frequency of the explicit referring to exploratory research rather than the frequency of exploratory research per se. The framing is then also different, for instance, underlining that it is crucial for the reader to being able to distinguish between exploratory and confirmatory research (as p-values have no clear meaning in exploratory research), to interpret a finding.

Author response: As noted above, we have addressed this major concern. We will note, however, that the reviewer is incorrect about O’Mahony’s findings. It is not true that the study “did or could not identify exploratory research in non-preregistered research.” Rather, they found a rate of 57% reported exploratory research in non-registered research.

(b) The abstract is a bit misleading. Citing from it: “reduce the frequency and value of exploratory research, and therefore restrict creativity, serendipity, and discovery” ...

“formally examine whether these concerns have any merit”. The use of “these” is misleading, as you do not examine “value of exploratory research, and therefore restrict creativity, serendipity, and discovery”, you only attempt to examine the frequency.

Author response: We have revised the abstract to clarify.

(2) Definition and protocol of exploratory research. On page 6 you define “... exploratory as, “any analysis that is indicated to not be a planned test of a specific hypothesis,””. Given this definition, I expected a protocol for ‘indicated not be a planned test of a specific hypothesis’. Please provide the protocol. For projects like this, and their reproducibility, a clear and objective protocol is essential. Without such a protocol (i) results will not be reproducible, and (ii) results are more difficult to interpret.

Author response: We now add a link to the full coding protocol. Importantly, there is no fully reproducible approach that can be used for this question. As we describe in the manuscript and in the manual, there is necessarily some subjectivity involved given that context must be taken into account. However, more clearly laid out coding procedures do indeed help with interpretation, and we will also make all of the articles and our coding openly available.

To make the protocol more objective and in line with my idea of the research (i.e., the explicit reference to exploratory research), I would use keywords like ‘plann*’ and ‘explor*’ and words that resemble it. This also will make coding easier and less subjective. Particularly because most published research does not list hypotheses or distinguish between planned and exploratory research.

Author response: Thank you, we have added several search terms in the coding protocol. As noted, this process will inherently involve some subjectivity.

(3) Hypotheses. Please explicitly list your planned hypotheses, also with “H1: ...”, “H2:...” etc. Improves the quality of your report, makes clearer what you are doing. We teach our students to do this, but we researchers should do it more in our research too.

Author response: Thank you, this point is well taken, and the hypotheses are now more clearly labeled at the end of the Introduction and when discussing the analyses.

Please also take care of your formulation on p7. It is not about “greater reports” but about the prevalence of exploratory research.

Author response: We agree that this formulation is more accurate and have edited the paper throughout.

Do you or do you not have a hypothesis about registered reports? This is not clear from your intro, as hypotheses are not explicitly listed. And do you have a hypothesis about comparing registered reports to ‘normal’ articles? It is not clear from the text if your hypothesis is of the form $A < B < C$, which implies three hypotheses on pairwise comparisons, or something else.

Author response: Thank you, we have reformulated the hypotheses to make our specific predictions clear.

Your exploratory hypotheses on p8 are also not clear to me. Do they concern the explicit use of exploratory in ALL published articles (i.e., unconditional) or in registered research (i.e., conditional)? Consider making it explicit.

Author response: The planned exploratory tests have now been clarified.

(4) Ideal registered report. For me, the ideal registered report is one where one simply adds the result section to an existing intro and methods section that is published in Stage 1. Although imo the introduction is quite good, it is not yet a full introduction of the paper (i.e., needs to be adapted). This depends on the journal, what are its criteria for a registered report.

Author response: We aim to keep the Introduction as brief as possible. We have revised it based on the comments, but otherwise have not added any additional information.

(5) What is a preregistration? On page 9 you define a preregistered paper as a paper with an “accessible link”. Please make this definition more meaningful. I can imagine accessible links to files that do not qualify as preregistrations.

Author response: We have revised the criteria to read as follows: “An empirical article in which a) all studies are stated to be preregistered and b) for which the registration document can be located (either via an included link in the manuscript or via manual search). Because of the wide variety in implementation quality, we will not require that the registration document has been formally registered, as long as it can be located (e.g., a registration document is on the OSF project page, but not registered).”

(6) Included studies. You wish to include any type of empirical studies, including qualitative, mixed methods, and meta-analyses. I understand, but (i) some designs (e.g., experimental) are more often preregistered than others, and (ii) results of these different designs may be different, and (iii) coding may be more cumbersome when dealing with more different designs. Because of these reasons I would understand if you focus on just ONE design, for instance, experimental designs. I think that this will also increase the statistical power of your design, as you control for possible confounding factors this way. If you want to stick to your original choice, then this is of course fine, but it would be great if you can motivate your choice more.

Author response: There is no reason to expect exploratory analyses would be more frequent in experimental vs. observational work, so we do not think it is important to restrict inclusion based on these types of designs. However, exploratory analyses could be different for qualitative, mixed methods, and meta-analytic studies, so we will now exclude these from the analysis.

(7) Article coding. You write on page 12 that ‘coding will be done at the article level rather than individual studies within articles’. To me, this sounds like an awful idea. I strongly recommend to code only the preregistered studies as preregistered, and not also the non-preregistered studies in the article. Actually, I think an excellent test of your hypothesis would be to compare preregistered studies and non-preregistered studies in the same article. Within-subject (in your case ‘within-article’) studies provide more control and more power for testing your hypotheses. You can then also compare if, within the same paper, authors distinguish more between exploratory and planned in preregistered studies than in non-preregistered studies.

Author response: As already indicated, we have altered the design to restrict the primary analysis to non-mixed articles and then conduct exploratory within-article analyses on the mixed articles.

(8) Rater training and reliability. Great that you include training. But why only six articles? Particularly if you include multiple different designs, six is not much imo. Why not 10 in the first stage? First one needs to develop a clear protocol, and then this protocol is tested in the first stage. Then, the protocol can be adapted, and a new batch can be coded, etc.

On page 15 you write “if agreement exceeds ... then the training phase will be complete”. I recommend not to include this rule, as it is an option that agreement never reaches this threshold. Most important is that you do your best with developing the protocol and training, until agreement cannot be improved much, independent of how much this agreement is.

You also write “Perfect agreement will be assessed...”. What does that mean?

Author response: In our many years of coding open-ended text and article data we have found that the detailed procedure works well, so we have stayed with the plan as described. We believe the reviewer may have misread the text, as the sentence in question reads “percent” not “perfect.”

(9) Statistical analyses I did not understand the statistical analyses (planned hypotheses). First, they seemed to be concerned with comparing frequencies, and then a chi2-test is an option. But then you write about ‘comparing averages’. What averages? And averages cannot be compared using a chi2-test. I had difficulties understanding this complete section.

Please make sure that you make clear how you test each of the explicitly tested hypotheses in the intro. Ideally, you also provide the statistical code for these tests.

Author response: This section has been completely overhauled, clearly indicating how each hypothesis is being tested, with links to the analytic code.

(10) Power analyses On page 12 you state that “the effect size in O’Mahony (2023) is not reported’. However, earlier in your report you provide the prevalences, with which you can calculate the effect size. The difference between 75 and 57 percent is not 20 percent, so this part confused me.

Author response: That sentence meant that the effect size phi was not reported, we have now edited the sentence for clarity. Regarding the percent difference, the sentence reads, “This corresponded to an approximate difference in prevalence of 20%” The actual difference was 18%, and in this sentence were providing an approximation. In any case, this sentence has been removed through revision of the power analysis.

Please make sure that the power analyses match the statistical tests used for testing the hypotheses (see (9) above).

Author response: This section has been completely overhauled, and the power analysis for each test is indicated after the test is described.

Perhaps it makes more sense to write the power analysis section following the planned analyses section, as you must describe the statistical analyses first. But perhaps the journal specified this order?

Author response: As noted, the power analysis for each test is now indicated following the description of the test.

Minor comments

“an accessible and” (p5): consider inserting “supposedly” between ‘a’ and ‘accessible’.

Author response: Changed

You use “1” twice on page 7 when listing statements. Note that it is sufficient to either explicitly state the H0 or the H1.

Author response: The section describing the hypotheses has been completely revised.

I would understand it if my review were perceived as unfavourable and undesired news. Sorry for that. However, I sincerely hope the authors can use my review to improve their research proposal, as more research is needed on this topic.

Keep up the good work!

Best wishes,

Marcel van Assen (I always sign my reviews)

Author response: We appreciate the feedback, and the transparency, and believe the comments have greatly improved the project.

PaperWizard

General Review Synopsis

This proposal addresses a timely metascience question but requires fundamental methodological refinement to yield valid inferences. Three critical issues warrant attention.

First, the operationalization measures reported exploratory research rather than its actual conduct, creating a construct validity problem that prevents adjudicating whether registration restricts exploratory work or merely clarifies its reporting. This distinction is central to addressing critics' concerns yet remains unresolved. Second, systematic measurement bias arises from detecting only explicitly labeled exploratory analyses—traditional articles likely contain unlabeled exploratory work, while registered articles face explicit labeling requirements, artificially inflating differences independent of actual practice changes. Third, the statistical approach requires revision: chi-square tests violate independence assumptions given journal-level clustering from the matching procedure, no corrections address multiple comparisons across 4-6 hypothesis tests, and the 2010 comparison group introduces temporal confounding that cannot be separated from registration effects. Addressing these through construct refinement, two-stage coding distinguishing labeled versus unlabeled exploration, mixed-effects models accounting for clustering, and temporal controls would substantially strengthen the study's capacity to provide definitive evidence for this important debate.

Author response: Each of these concerns is addressed below when raised.

Novelty Assessment

Novel - The proposal presents empirical novelty by systematically testing a long-debated claim that preregistration/registered reports restrict exploratory research. Despite extensive searches, no published papers were found that directly compare the frequency of reported exploratory analyses across registered vs. traditional articles in psychology. The authors cite O'Mahony (2023) as the only relevant data, but this is an unpublished PhD thesis (<https://orca.cardiff.ac.uk/id/eprint/167686>) that examined this question within a larger project. The current proposal substantially extends this by: 1) including preregistered articles (not just registered reports), 2) adding historical comparison (2010 articles), 3) using systematic matching criteria, 4) employing transparent coding procedures with reliability assessment, and 5) explicitly testing temporal trends. The proposal provides the first peer-reviewed, systematic empirical test of whether registration practices actually suppress exploratory research reporting as critics claim, moving the debate from opinion to data. The research addresses a scientifically and practically important question about open science reform impacts with a rigorous design that will provide definitive empirical evidence currently absent from the published literature. For details on the most similar publications, see Appendix 1.

Author response: Yes, these are indeed strengths of the project.

Issues

1.1 Construct Validity and Operationalization

Alignment Between Measurement and Research Question -

Critique: The study's core limitation—measuring **reported** exploratory research rather than its actual conduct—requires careful consideration regarding whether the design can meaningfully address critics' concerns about registration **restricting** exploratory work

(Nosek, 2018)(Goldin-Meadow, 2016). Critics argue registration constrains research practices ("putting researchers in chains," "stifling discovery"), yet the operationalization conflates reporting transparency with behavioral impact. Observing Pattern 2 (more reported exploration in registered articles) could reflect improved labeling rather than unrestricted conduct, while Pattern 1 might indicate restriction or merely poorer reporting. This ambiguity prevents adjudicating between registration enabling exploration, clarifying it, or restricting it while enhancing visibility of remaining work.

Relevant section: Abstract; "The Present Study" section, particularly the acknowledgment that "the current study will not be able to actually assess the frequency of exploratory research, per se, but rather the frequency with which exploratory research is reported and transparently described" (p.5).

Framework for Addressing: - Reframe research questions explicitly in terms of reported exploratory research transparency rather than actual conduct restriction (e.g., "Is exploratory research more transparently reported in registered versus traditional articles?")

- Add a dedicated limitations subsection explicitly discussing how findings about reporting practices do/do not address critics' concerns about practice restriction
- Develop supplementary analyses that could more directly assess conduct (e.g., examining preregistration-manuscript discrepancies to identify unreported exploratory work in registered articles)
- Specify in the Discussion how different patterns would be interpreted: finding more reported exploration in registered articles supports transparency claims but does not refute restriction of actual conduct
- Consider adding author surveys or interviews with a subsample to directly assess whether registration affected their exploratory research behavior versus only their reporting

Author response: These issues have already been addressed in response to previous comments. Adding an author survey/interviews is a good idea, but is not possible given the restricted timeline for the project.

Systematic Detection Bias in Traditional Articles -

Critique: The operational definition introduces measurement error systematically favoring the hypothesis by detecting only explicitly labeled exploratory research. Traditional articles written before widespread preregistration norms (particularly 2010 articles) likely contain unlabeled exploratory analyses because authors had no reason to distinguish confirmatory from exploratory work (Wagenmakers et al., 2012). Conversely, registered articles face explicit requirements (Registered Report formats mandate separate labeling; preregistration badges require this distinction) to flag exploratory work. This creates artifactual differences: registered articles will have exploratory work explicitly labeled per journal requirements, while traditional articles may contain equivalent exploration remaining undetected. The study cannot distinguish whether registered articles contain MORE exploratory work or simply LABEL it more, directly affecting the ability to test competing explanations. Relevant section: Method section, "Statements of Exploratory Research" coding criteria (pp.10-11).

Framework for Addressing: - Implement a two-stage coding process: first, identify explicitly labeled exploratory research; second, conduct detailed analysis plan comparisons

(for registered articles with accessible registrations) to identify conducted-but-unlabeled exploratory analyses

- Add systematic searches for common exploratory analyses that may be unlabeled in traditional articles (e.g., post-hoc subgroup analyses, alternative model specifications, additional dependent variables not mentioned in introduction)
- Code separately: (a) explicitly labeled exploratory research and (b) analyses meeting the definitional criteria but unlabeled, then analyze these as distinct outcomes
- Include sensitivity analyses comparing only registered articles with accessible, detailed preregistrations to traditional articles, where unlabeled exploration can be more reliably detected through plan-manuscript comparison
- Report both "any exploratory research present" and "exploratory research explicitly labeled as such" as separate outcomes to distinguish transparency from conduct

Author response: As noted, we are now clearer that we are focusing on reporting, which is the only possible way to assess the research question. Given widespread HARKing, there is no way to know the "true" rate of exploration outside of registration constraints. The comments are also not correct, as we are analyzing both explicit and implied exploratory work.

Operationalization of Context-Dependent Exclusions -

Critique: The protocol's reliance on subjective "context" for excluding certain analyses (interaction probing, exploratory factor analysis) introduces inconsistency without objective specification. The authors cite (Baranger et al., 2023) acknowledging interaction hypotheses typically lack specificity, yet exclude interaction probing when it "follows a hypothesized interaction"—contradicting recognition that non-specific interaction predictions render follow-up probing exploratory by the study's definition ("any analysis not a planned test of a specific hypothesis"). Without objective criteria, raters may systematically judge interaction probing as confirmatory in registered articles (where interactions were preregistered) but exploratory in traditional articles, even when hypothesis specificity is equivalent.

Relevant section: "Statements of Exploratory Research" (pp.10-11), particularly the paragraph discussing "not all uses of the word 'explore' or 'exploratory' necessarily indicate exploratory research."

Framework for Addressing: - Provide explicit, objective criteria for context-dependent decisions:

"Interaction probing is exploratory unless (a) the preregistration/introduction specifies the directional pattern of simple effects, AND (b) the analysis matches this specification exactly"

- Develop detailed decision trees with specific examples for each ambiguous category (interaction probing, EFA, robustness tests) showing when each would/would not be coded as exploratory
- Include these decision rules in the codebook provided during training with 5-10 worked examples per category
- Track and report separately the frequency of context-dependent decisions by article type to assess whether systematic biases emerge

- Consider including all interaction probing consistently as exploratory unless extraordinarily specific a priori predictions are documented, given Baranger et al.'s demonstration of typical non-specificity

Author response: This has already been addressed in response to previous comments, we now provide a more detailed codebook with coding criteria.

Construct Representation Through Binary Coding -

Critique: Binary presence/absence coding across three sections, averaged to create 0-1 scores, severely underrepresents the construct and cannot detect meaningful restriction. Two articles receiving identical scores could differ dramatically: one reporting a single footnote robustness check versus another with extensive exploratory analyses comprising 50% of Results driving primary conclusions. If critics' concerns about "putting researchers in chains" have merit, effects would likely manifest in reduced proportion of results devoted to exploration, less prominent placement, fewer exploratory findings driving conclusions, or reduced sophistication—dimensions this coding cannot capture. The sensitivity analysis suggesting detection of $d=0.35$ misleadingly assumes binary coding adequately represents the construct, when meaningful restriction could exist with "present" exploration that is perfunctory or minimized.

Relevant section: "Article Coding" section (pp.9-11), particularly "Statements of Exploratory Research."

Framework for Addressing: - Expand coding to capture: (1) count of distinct exploratory analyses reported, (2) estimated proportion of Results section text devoted to exploratory findings (categorical: 0%, 1-10%, 11-25%, 26-50%, >50%), (3) whether exploratory findings contribute to primary conclusions (yes/no), (4) prominence of exploratory work (main text vs. supplement/footnote)

- Analyze both binary presence/absence and continuous/ordinal measures of extent, comparing patterns across both operationalizations
- Report effect sizes for both "any exploratory research" and "substantial exploratory research" (e.g., >25% of results) to distinguish restriction from minimization
- Conduct sensitivity analyses examining whether patterns differ when restricting to articles with substantial versus minimal exploratory work
- Example expansion: "Code proportion of Results devoted to exploratory analyses: 0 (none), 1 (1-10%, typically single robustness check), 2 (11-25%, multiple supplementary analyses), 3 (26-50%, major results component), 4 (>50%, primary focus)"

Author response: To address this concern, we now distinguish between the presence of exploratory research (binary yes/no) and prominence of exploratory research (count 0-6) based on ratings of different sections of the articles.

1.2 Article Selection and Matching Procedures

Matching Hierarchy and Comparability -

Critique: The hierarchical matching prioritizing Timeframe over Topic/Design similarity requires reconsideration, as accepting Topic/Design ratings of 3 ("not similar") to maintain Timeframe ☐ 2 pairs fundamentally incomparable articles (e.g., depression interventions with cognitive neuroscience studies) from adjacent issues. This approach conflates

registration effects with unmeasured confounds including statistical complexity, sample size requirements, field-specific norms regarding exploratory analysis, and disciplinary variations (experimental vs. observational traditions). The subjective 3-point ratings lack demonstrated reliability, no empirical justification supports prioritizing temporal proximity over substantive similarity, and no verification ensures balanced score distributions or sensitivity analyses excluding poor matches. Such mismatches could bias differences toward artifacts of topic clustering rather than registration status, particularly given that research areas adopting registration may differ systematically in complexity and analytic practices (Claesen et al., 2021).

Relevant section: "Article Selection Process" (pp.6-9), specifically the "Companion articles will be selected based on the following criteria" subsection describing matching hierarchy.

Framework for Addressing: - Reverse the hierarchy to prioritize Topic ≥ 2 and Design ≥ 2 as mandatory thresholds, then minimize Timeframe within these constraints

- Establish objective criteria for Topic/Design ratings with examples: Topic 1="both examine depression interventions," 2="both examine psychopathology," 3="both clinical but different domains"; provide similar specifications for Design
- Conduct inter-rater reliability assessment on Topic/Design ratings using 50 pilot articles, requiring $\kappa \geq .70$ before applying to full sample - Report distributions of matching quality scores and conduct sensitivity analyses: (a) excluding matches with Topic or Design=3, (b) including matching quality as covariates in analyses
- Add statistical controls for measured confounds: code article complexity (number of studies, analyses, sample size categories) and research domain (experimental, observational, clinical, social, cognitive), include as covariates
- Example addition to matching criteria: "Companion articles must achieve minimum scores of 2 for both Topic and Design. Within this constraint, select the article with the closest publication date. If multiple articles meet all criteria, prioritize Topic similarity (1>2) over Design similarity."

Author response: We disagree with this assessment. Given the shifting norms and practices at journals (see MetaROR reviewer comments), we continue to assert that timeframe is the most pertinent matching criterion.

Selection Bias From Sampling Frame -

Critique: The sampling frame introduces fundamental selection bias that cannot be separated from registration effects. Starting exclusively with 47 journals publishing Registered Reports means comparing traditional articles in journals that adopted RRs against traditional articles in those same journals. Journals adopting RRs likely possess pre-existing editorial cultures valuing transparency, exploratory research, and open science differently than psychology journals generally (Chambers & Tzavella, 2020). Any observed differences could reflect journal selection effects rather than registration effects—higher exploratory rates in registered articles might indicate (a) registration clarifying exploration, (b) RR-publishing journals attracting authors valuing exploration, or (c) editorial cultures encouraging exploratory reporting regardless of registration. The design provides no mechanism to distinguish these explanations.

Relevant section: "Article Selection Process" (pp.6-9), specifically the rationale "These journals are a good starting point, because publishing Registered Reports is a strong signal that the journal values registration."

Framework for Addressing: - Add a comparison group: sample traditional articles from journals that do not publish Registered Reports but match on impact factor, subdiscipline, and editorial scope to assess whether traditional articles in RR-publishing journals are representative

- Conduct supplementary analyses comparing traditional articles in RR-publishing journals to this matched control group from non-RR journals

- If adding this comparison is infeasible, acknowledge this as a critical limitation: "Our findings compare registered and traditional articles within journals adopting Registered Reports; results may not generalize to psychology broadly if these journals have distinctive editorial cultures"

- Code and analyze journal-level characteristics (e.g., open science policies, impact factor, publication volume) as potential moderators

- Frame conclusions specifically: "Within journals that publish Registered Reports..." rather than making general claims about registration's impact on psychology

Author response: As already noted in response to the human reviewers, we are a) analyzing journal policies and b) including a small comparison sample of traditional articles published in journals that do not publish Registered Reports.

Treatment of Multi-Study Articles and Unit of Analysis -

Critique: Including articles where only one study among multiple is registered while coding exploratory research article-wide introduces misclassification bias that obscures registration effects. Articles classified as "registered" may contain substantial exploratory work in non-registered studies, making them functionally indistinguishable from traditional articles regarding the non-registered portions. The justification—that researchers using mixed formats to include both confirmatory and exploratory analyses would be "consistent with registration advocates' position"—embeds the conclusion being tested into the methodological choice, creating circular reasoning. More critically, this decision misaligns measurement with the theoretical level at which registration operates: registrations apply to specific studies, not entire articles. If authors strategically segregate exploration to non-registered studies within mixed articles, this would support rather than refute restriction concerns.

Relevant section: "Article Coding" (pp.9-10), particularly the paragraph stating "Coding will be done at the article-level rather than individual studies within articles."

Framework for Addressing: - Restructure to study-level analysis: for multi-study articles, code each study separately, recording whether that specific study was registered and the exploratory research within that study's corresponding results

- Compare registered studies to non-registered studies both between and within articles, allowing detection of whether authors segregate exploration by registration status

- If article-level coding is retained, restrict inclusion to articles where ALL studies share the same registration status (all registered or all non-registered) to ensure clean categorization

- Conduct sensitivity analyses comparing results when including versus excluding mixed-registration articles

- Code registration proportion for multi-study articles (e.g., 1 of 3 studies registered = 33%) and examine this as a continuous moderator to test dose-response relationships
- Report separately: analyses of pure-type articles only (all studies registered/non-registered) versus all articles including mixed types, with explicit comparison of conclusions

Author response: This was already addressed in response to previous comments, we are now excluding mixed articles from the primary analysis but will conduct exploratory within-article comparisons on the set that we identify.

1.3 Coding Reliability and Measurement Quality

Adequacy of Reliability Thresholds -

Critique: The target thresholds (80% agreement, $\kappa \geq .60$) require reconsideration given the subjective judgments required and the study's validity dependence on accurate identification. Kappa of .60 represents "moderate agreement" by conventional standards (Landis & Koch, 1977), yet coding involves inferring author intent through "clearly implied" meanings where exploratory research "lacks clear or consistent definition." In content analysis requiring subjective interpretation, standards typically mandate $\kappa \geq .70$ -80 for acceptable reliability (Hallgren, 2012). Additionally, 80% agreement could mask systematic disagreements—if two raters consistently disagree on 20% of articles in the same direction (one systematically more lenient), this meets the threshold while introducing substantial bias. Given the hypothesized moderate effect ($\phi = .18$), measurement error at this level could produce spurious findings or obscure true effects.

Relevant section: "Rater Training and Reliability" (pp.11-12), specifically "If percent agreement exceeds .80 and Kappa exceeds .60, then the training phase will be complete."

Framework for Addressing: - Increase thresholds to 85% agreement and $\kappa \geq .75$ before proceeding to focal coding, consistent with standards for subjective content analysis requiring inference about author intent

- Examine not only overall agreement but also direction of disagreements: calculate separate agreement for "exploratory present" versus "exploratory absent" decisions to detect systematic leniency/strictness biases
- Track disagreement patterns by article type during training to identify whether certain article types (e.g., traditional) are systematically more difficult to code - If increased thresholds cannot be achieved after extended training, conduct all focal coding with three raters per article (instead of two), using majority rule or requiring consensus
- Report reliability metrics separately for each section (Introduction, Results, Discussion) and each article type, as reliability may vary across these dimensions
- Example addition: "During training, we will examine whether disagreements are randomly distributed or systematically favor certain coding decisions (e.g., one rater coding ambiguous cases as exploratory more often), adjusting training protocols if systematic biases emerge."

Author response: Kappa assumes that the marginal distributions are equal. Violations of this lead to lower values, which is a known issue. We will almost certainly observe unequal marginals, hence the lower threshold.

Blinking Procedures and Expectancy Effects -

Critique: The absence of blinding procedures in coding where raters identify "clearly implied" exploratory research requires attention, as knowledge of article type may introduce expectancy effects. Registered Reports' distinctive two-stage structure and preregistration statements make article types readily identifiable, potentially leading raters to apply differential thresholds when inferring unlabeled exploration: stricter thresholds for registered formats (expecting explicit labels per requirements) and looser ones for traditional articles. Training achieving only moderate reliability ($\kappa=.60$) provides insufficient safeguard against shared biases among the four-person team who collaborate weekly. No procedures for partial blinding (e.g., redacting registration information, removing Stage 1/2 headers), bias assessment (testing whether raters can identify article types and whether this affects coding), or differential error checks (comparing disagreement rates by article type) are specified.

Relevant section: "Rater Training and Reliability" (pp.11-12) and "Article Coding" (pp.9-11); refers to omissions throughout.

Framework for Addressing: - Implement partial blinding: create redacted versions removing obvious registration indicators (preregistration badges, Stage 1/Stage 2 section headers, explicit registration statements, OSF links) for a subset of articles

- Randomly assign 100 articles to blinded condition (redacted) and 200 to unblinded condition; compare coding patterns across conditions to assess bias magnitude
- Add explicit bias checks: after completing 50% of coding, have raters independently guess each article's type; calculate identification accuracy and test whether articles where type was correctly identified show different coding patterns
- Include "confidence in coding decision" ratings (1-5 scale) to identify cases where rater uncertainty might reflect ambiguity rather than true construct presence
- Analyze inter-rater reliability separately by article type; substantially lower reliability for one type would indicate differential measurement quality requiring investigation
- Example: "To assess potential expectancy effects, we will redact registration-identifying information from 100 randomly selected articles and compare coding reliability and outcomes between redacted and non-redacted versions."

Author response: Blinding articles in the way suggested is not possible without removing other key indicators that are important for assessing the articles, see Soderberg et al. (2021) for a discussion of this issue. Therefore, we will include this only as a limitation.

1.4 Statistical Approach and Power

Accounting for Nested Data Structure -

Critique: The planned chi-square tests require reconsideration given multiple non-independence sources violating the tests' assumptions. Articles are explicitly matched by journal (creating clustering within journals sharing editorial policies, reviewer pools, and cultures), may originate from identical research groups, and exhibit temporal clustering (same/adjacent issues). The matching procedure intentionally creates dependencies: companion articles selected from identical journals and proximate issues means observations are nested within journals and time periods. Chi-square assumes independent observations; these violations will inflate Type I error rates and bias effect size

estimates (McNeish & Stapleton, 2016). The analysis plan mentions neither hierarchical/multilevel modeling to accommodate clustering, nor sensitivity analyses for independence violations, nor corrections for non-independence.

Relevant section: "Planned Analysis" (pp.12-13), particularly "Hypothesis Tests" subsection specifying "2x2 chi-square test of independence ($\alpha = .05$)" and "3x3 chi-square test of independence."

Framework for Addressing: - Replace chi-square tests with generalized linear mixed models (GLMMs) specifying random intercepts for journal and potentially time period (publication year or issue number)

- For binary outcomes (exploratory research present/absent), use mixed-effects logistic regression; for averaged ratings, use linear mixed models
- Report intraclass correlation coefficients (ICCs) quantifying the proportion of variance attributable to journal-level versus article-level factors
- Conduct sensitivity analyses comparing naive chi-square results to mixed model results to demonstrate the impact of accounting for clustering
- If sample size per cluster is small (few articles per journal), consider cluster-robust standard errors as alternative approach
- Example model specification: "We will fit a mixed-effects logistic regression predicting exploratory research presence from article type (RR, preregistered, traditional), with random intercepts for journal to account for clustering. Fixed effects will be tested using Wald tests with Kenward-Roger degrees of freedom adjustment."

Author response: This is a good suggestion. Given the very small number of likely articles spread across 47 journals, we opted to use chi-square and t-tests that adjust for clustering, rather than specifying random effects in mixed-effects models. We will also conduct sensitivity tests by conducting "naive" tests that do not account for clustering.

Multiple Comparisons and Family-Wise Error -

Critique: The analysis plan requires addressing multiple comparisons despite planning numerous hypothesis tests, which substantially inflates family-wise Type I error rate. Planned analyses include: (1) 2x2 chi-square for H1, (2) "secondary test" examining headers for H1, (3) 3x3 chi-square for H2, (4) "secondary test" examining headers for H2, and (5) follow-up cell-wise residual analyses for significant results using ± 1.96 thresholds—minimally 4-6 inferential tests plus additional follow-ups if omnibus tests achieve significance. The "primary" versus "secondary" distinction lacks statistical meaning for family-wise error control; both contribute to inflation.

With 4-6 planned tests at $\alpha = .05$, probability of ≥ 1 Type I error under the null substantially exceeds the nominal .05 level. The proposal's emphasis on providing "strong data" for policy discussions heightens the importance of controlling false positive rates (Rothman, 1990) (Benjamini & Hochberg, 1995).

Relevant section: "Planned Analysis" (pp.12-13), "Hypothesis Tests" subsection; refers to omission of multiple comparison corrections throughout.

Framework for Addressing: - Specify a priori which tests constitute the primary confirmatory family requiring correction (minimally: the four hypothesis tests for H1 and H2, both primary and secondary)

- Apply Holm-Bonferroni sequential procedure or FDR correction (Benjamini-Hochberg) to this family, adjusting the decision rule accordingly
- For follow-up tests (cell-wise residuals), apply correction within each omnibus test family or use more stringent thresholds (e.g., ± 2.58 for $p < .01$)
- Distinguish clearly between confirmatory tests (corrected α) and truly exploratory analyses (reported without p-values as planned)
- Report both uncorrected and corrected results for transparency, but base conclusions on corrected values
- Example: "We will control family-wise error rate across the four primary hypothesis tests using the Holm-Bonferroni method. For H1 and H2 to be considered supported, both the primary and secondary tests must achieve significance at the adjusted α levels. Follow-up analyses will be conducted only for omnibus tests surviving correction."

Author response: We now specify that we will use the Benjamini Hochberg false discovery rate for error control.

1.5 Confounds and Alternative Explanations

Temporal Confounding With Historical Comparison Group -

Critique: The inclusion of 75 traditional articles from 2010 "prior to registration adoption" introduces temporal confounding that compromises primary analyses. Comparing recent registered articles (2015-2026) to a mixed pool including 2010 articles creates ambiguity: observed differences could reflect registration effects, 15-year secular changes in reporting norms, or their interaction. This period witnessed substantial evolution: the replication crisis gained prominence (2011-2012), transparency initiatives proliferated, APA reporting standards updated (2010, 2013, 2020), and journal policies regarding open science badges and data sharing increased dramatically—affecting traditional articles independently of preregistration. The power analysis assumes simple comparisons between contemporaneous groups, but the design involves more complex structure where one category spans different (and longer) timeframes. The current approach confounds factors in primary analyses while relegating time investigation to "exploratory" analyses without inferential tests.

Relevant section: "Article Selection Process" (pp.6-9), particularly the paragraph describing "a second group of articles published in 2010, prior to the adoption of registration in psychology." **Framework for Addressing:** - Restrict all articles to the same time window (2015-2026 only) for primary hypothesis tests, ensuring temporal comparability

- Analyze 2010 articles separately as a distinct historical baseline, not combined with matched traditional articles in primary analyses
- If 2010 articles are retained, explicitly model time as a factor: use $2 \times 2 \times 2$ design (Article Type: Registered/Traditional $\times$ Time Period: 2010/2015-2026) allowing separation of registration and temporal effects
- Test registration $\times$ time interaction to determine whether registration's association with exploratory reporting differs between periods
- Report primary analyses both with and without 2010 articles, clearly labeling which constitute the definitive hypothesis tests

- Example revision: "Primary analyses will compare 75 Registered Reports, 75 preregistered articles, and 75 matched traditional articles all from 2015-2026. Separately, we will compare the 75 matched traditional articles (2015-2026) to 75 traditional articles from 2010 to assess temporal changes in reporting practices independent of registration."

Author response: The 2010 articles will only be included for comparison to address the exact concern raised here about changes over time, and will not be included in the primary analyses comparing the article types. This has now been clarified in the manuscript.

Registration Quality as Unmeasured Confounder -

Critique: The decision to remain "agnostic about registration quality" constitutes a critical design limitation given documented variation in registration specificity. Multiple studies demonstrate enormous quality heterogeneity (van den Akker et al., 2023), with poor-quality registrations lacking specificity functionally rendering all analyses exploratory by default—artificially inflating exploratory reporting in registered articles. Conversely, high-quality registrations with specific predictions might show reduced exploratory work as critics predict, but this effect would be masked by including poor registrations in the same category. The authors acknowledge this variation exists but dismiss it as "not our concern here"—yet if registration quality is the actual causal mechanism (only high-quality registrations potentially restricting exploration), ignoring it violates the requirement of measuring treatment actually delivered. The power analysis assumes homogeneous treatment effects; unmeasured heterogeneity could mean the study lacks power to detect effects within quality strata.

Relevant section: "Article Coding" (pp.9-11), paragraph stating "in the current project we remain agnostic about the quality of the registrations themselves."

Framework for Addressing: - Code registration specificity for all registered articles using established frameworks: specify whether hypotheses, predictions, analysis plans, and exclusion criteria are clearly detailed (binary or 4-point scale per element)

- Calculate overall registration quality score (e.g., sum across elements, range 0-4 or 0-8) and categorize as high/medium/low quality using quartiles or established cutoffs

- Conduct analyses stratified by registration quality: test whether associations between registration and exploratory reporting differ across quality levels

- Report separate effect estimates for high-quality versus low-quality registrations to determine if patterns are quality-dependent

- Test registration quality as a moderator using interaction terms in statistical models

- Example addition: "We will assess registration quality by coding four elements: (1) specificity of hypotheses/predictions, (2) detail of planned analyses, (3) specification of exclusion criteria, (4) clarity of outcomes (each rated 0-2). We will examine whether associations between registration status and exploratory reporting vary by total quality score (0-8), testing whether effects are driven by high-quality registrations only."

Author response: As already noted in response to the human reviewer, we will not be adding analyses of the preregistered protocol, but will include that as a limitation and avenue for future directions when writing up the final report.

ReviewerZero

The only “issue” identified by this service was that the Abstract showed 99.6% overlap...but this overlap was with our own abstract for the project that had been auto-uploaded to Research Gate. Thus, ReviewerZero did not identify any issues that required action.